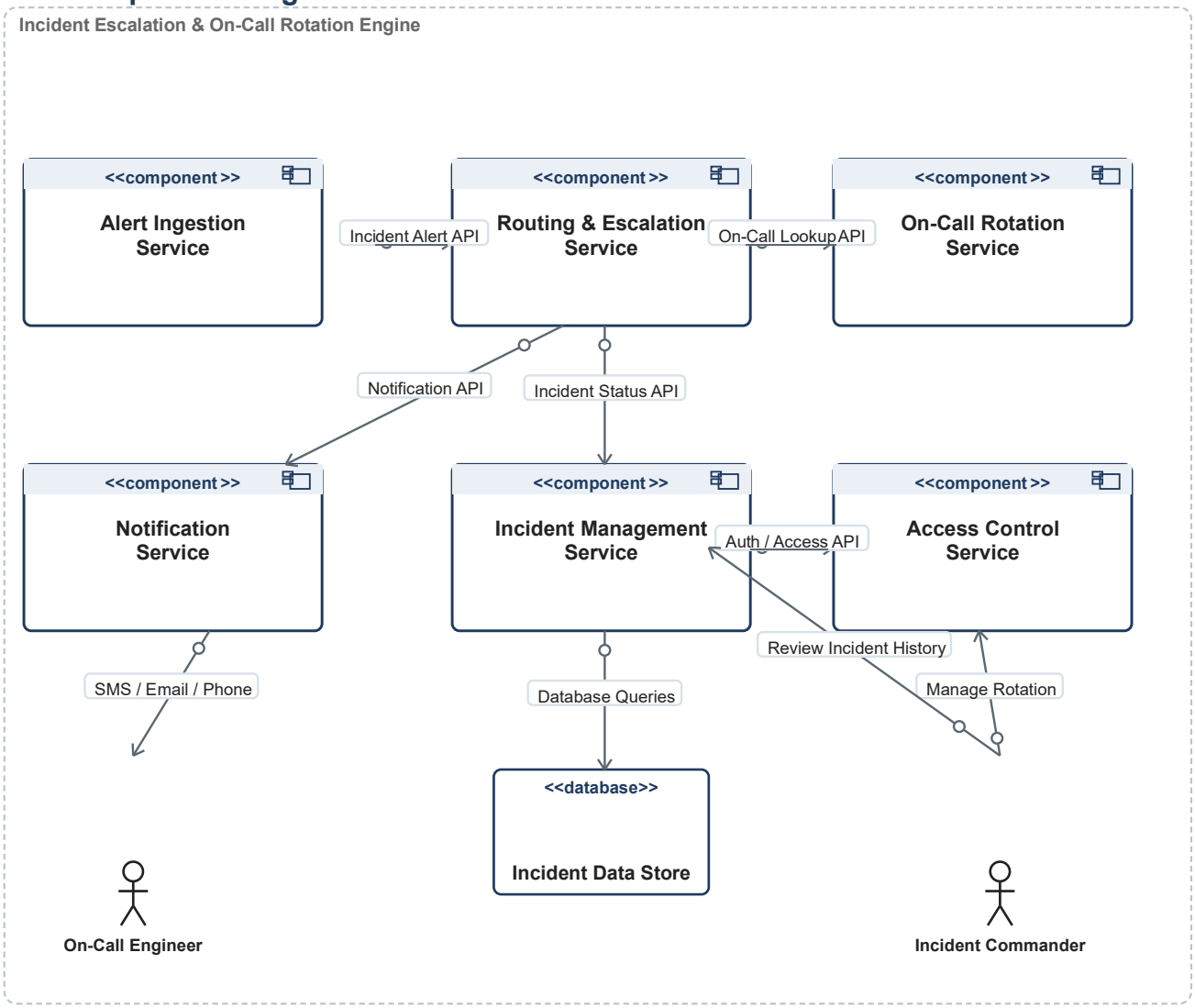
LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Incident Escalation & On-Call Rotation Engine

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Scenario	48	**Selected Architecture**	Microservices Architecture

UML Component Diagram



Circle on connector = provided interface; arrow points toward the component using/receiving the service.

Figure 1. Component diagram for Scenario 48. The model separates alert ingestion, routing/escalation, rotation management, notification, incident records and access control, with labelled service interfaces.

ARCHITECTURAL STYLE JUSTIFICATION

Incident Escalation & On-Call Rotation Engine

Architecture Selection

We chose Microservices Architecture for the Incident Escalation & On-Call Rotation Engine. The system naturally separates into services for alert ingestion, routing and escalation, on-call rotation, notifications, incident records and access control.

Reason 1 - Separation of Responsibilities

The functional requirements describe distinct responsibilities: receiving alerts and identifying the active engineer, sending notifications, escalating an unacknowledged incident after 5 minutes, managing primary and secondary rotation schedules, and storing resolution/post-mortem information. Keeping these responsibilities as separate services makes each part easier to develop, test and update without changing the whole system.

Reason 2 - Reliability and Fault Isolation

Incident response is time-sensitive, so one failing function should not stop the entire workflow. In a microservices design, a failure in a service such as notification can be isolated while routing, rotation lookup and incident recording remain independently recoverable. This is appropriate for an operational system where missed incidents can have a high impact.

Security Advantage

The scenario requires only authenticated and authorized users to access incident data, rotation schedules and escalation settings. A dedicated access-control service can enforce authentication and role-based authorization at service boundaries, limiting sensitive operations to permitted users such as the On-Call Engineer and Incident Commander.

Performance Benefit

The performance requirement states that alert dispatch through webhook, SMS and email should begin within 3 seconds of alert ingestion. Microservices allow high-demand parts such as Alert Ingestion and Notification to be scaled independently, helping the system handle peak incident volume without scaling every component.

Component	Responsibility
Alert Ingestion Service	Receives and records incoming monitoring alerts.
Routing & Escalation Service	Assigns the active engineer and triggers the 5-minute escalation rule.
On-Call Rotation Service	Maintains and provides primary/secondary on-call schedules.
Notification Service	Sends SMS, phone or email notifications.
Incident Management Service	Stores status, resolution details and post-mortem information.
Access Control Service	Enforces authentication and authorization.
Incident Data Store	Persists incident, schedule and post-mortem data.